AQ3.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

- [ ] The set $\{(1,2,3),(4,5,6),(7,8,9)\}$ is linearly dependent.
- [ ] The set $\{(1,2,3),(4,5,6),(5,7,9)\}$ is linearly independent.
- [ ] The set $\{(1,2,3),(0,5,6),(0,0,9)\}$ is linearly independent.
- [ ] The set $\{(1,2,3),(0,0,6),(0,0,9)\}$ is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1,2,3),(4,5,6),(7,8,9)\}$ is linearly dependent.

The set $\{(1,2,3),(0,5,6),(0,0,9)\}$ is linearly independent.

1 point

If $ad - bc = 0$ then which of the following options are true?

- [ ] The set $\{(a,b),(c,d)\}$ is linearly dependent in R^2.
- [ ] The set $\{(a,c),(b,d)\}$ is linearly dependent in R^2.
- [ ] The set $\{(1,0,0),(0,a,b),(0,c,d)\}$ is linearly dependent in R^3.
- [ ] The set $\{(1,0,0),(0,a,c),(0,b,d)\}$ is linearly dependent in R^3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(a,b),(c,d)\}$ is linearly dependent in R^2.

The set $\{(a,c),(b,d)\}$ is linearly dependent in R^2.

The set $\{(1,0,0),(0,a,b),(0,c,d)\}$ is linearly dependent in R^3.

The set $\{(1,0,0),(0,a,c),(0,b,d)\}$ is linearly dependent in R^3.

For $c \in Z$, consider the set of three vectors $S = \{(1,c,0),(c,0,c),(2c,3c,4c)\}$ in R^3 with usual addition and scalar multiplication. Find the value of c such that the given set is linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

1 point

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly independent and consider the following system of linear equations

$$a_1 x + b_1 y + c_1 z = 1$$
$$a_2 x + b_2 y + c_2 z = 2$$
$$a_3 x + b_2 y + c_3 z = 3.$$

z = 3.

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then which of the following option(s) is(are) true?

- ☐ The system of linear equations has a unique solution.
- ☐ The system of linear equations has no solution.
- ☐ The system of linear equations have infinitely many solutions.
- ☐ The coefficient matrix A is invertible.
- ☐ The coefficient matrix A is not invertible.
- ☐ Any one of the above options cannot be concluded using the given information.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The system of linear equations has a unique solution.
The coefficient matrix A is invertible.

1 point

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$ is given to be linearly dependent and consider the following system of linear equations

$$a_1 x + b_1 y + c_1 z = 1$$
$$a_2 x + b_2 y + c_2 z = 2$$
$$a_3 x + b_2 y + c_3 z = 3.$$

z = 3.

If $AX = b$ is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then which of the following option(s) is(are) true?

- ☐ Either the system of linear equations has a unique solution or no solution.
- ☐ Either the system of linear equations has no solution or infinitely many solutions.
- ☐ Either the system of linear equations have infinitely many solutions or a unique solution.
- ☐ The system of linear equations has a unique solution.
- ☐ The coefficient matrix A is invertible.
- ☐ The coefficient matrix A is not invertible.

No, the answer is incorrect.

Score: 0

Accepted Answers:
Either the system of linear equations has no solution or infinitely many solutions.
The coefficient matrix AA is not invertible.
If SS is a linearly independent set in vector space R^3R3 with respect to usual addition and scalar multiplication, then find the maximum possible cardinality of SS.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Level 2

1 point
Let the set $\{(a,b),(c,d)\}$\{(a, b), (c, d)\} be a linearly independent subset of R^2R2. Choose the set of correct options.

- [ ] $\{(a,b,0),(c,d,0)\}$\{(a, b, 0), (c, d, 0)\} must be a linearly independent subset of R^3R3.
- [ ] $\{(a,b,0),(c,d,0),(0,0,1)\}$\{(a, b, 0), (c, d, 0), (0, 0, 1)\} must be a linearly independent subset of R^3R3.
- [ ] $\{(a,b,0),(c,d,0),(1,0,0)\}$\{(a, b, 0), (c, d, 0), (1, 0, 0)\} must be a linearly independent subset of R^3R3.
- [ ] $\{(a,b,0),(c,d,0),(0,1,0)\}$\{(a, b, 0), (c, d, 0), (0, 1, 0)\} must be a linearly independent subset of R^3R3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(a,b,0),(c,d,0)\}$\{(a, b, 0), (c, d, 0)\} must be a linearly independent subset of R^3R3.
$\{(a,b,0),(c,d,0),(0,0,1)\}$\{(a, b, 0), (c, d, 0), (0, 0, 1)\} must be a linearly independent subset of R^3R3.
1 point
Each column of an $n \times n$n × n matrix AA can be thought of as vectors in R^nRn. If a set SS consists of all the column vectors of AA, then which of the following is(are) true?

- [ ] SS is always linearly independent.
- [ ] SS is linearly independent if $det(A) \neq 0$det(A) = 0.
- [ ] SS is linearly independent if $det(A) = 0$det(A) = 0.
- [ ] If $det(A) = 1$det(A) = 1, then SS is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
SS is linearly independent if $det(A) \neq 0$det(A) = 0.
If $det(A) = 1$det(A) = 1, then SS is linearly independent.
1 point

Suppose it is known that the three vectors $v_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, v_2 = \begin{bmatrix} d \\ e \\ f \end{bmatrix}, v_3 = \begin{bmatrix} g \\ h \\ i \end{bmatrix}$v1 = $\begin{bmatrix} \\ \\ \\ \end{bmatrix}$abc , v2 = $\begin{bmatrix} \\ \\ \\ \end{bmatrix}$def , v3 = $\begin{bmatrix} \\ \\ \\ \end{bmatrix}$ghi are linearly dependent in R^3 R3. Which of the following is correct?

- [ ] Some linear combination of a, b, c a, b, c must be 0.
- [ ] Some linear combination of a, d, g a, d, g must be 0.
- [ ] Some linear combination of d, e, f d, e, f must be 0.
- [ ] Some linear combination of c, f, i c, f, i must be 0.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Some linear combination of a, b, c a, b, c must be 0.
Some linear combination of a, d, g a, d, g must be 0.
Some linear combination of d, e, f d, e, f must be 0.
Some linear combination of c, f, i c, f, i must be 0.

If vectors $(2x, 4, -6), (-6, -3x, 2x), (-1, x-3, x+2)$(2x, 4, −6), (−6, −3x, 2x), (−1, x − 3, x + 2) from the vector space R^3 R3 with usual addition and scalar multiplication, are linearly dependent and $x \in [2, 5]$x ∈ [2, 5], then find the value of xx.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point